

Master Spoken Voiceover Narration Script

Complete Voice Lines for Text-to-Speech (TTS) & Studio Voiceover Production

TOTAL AUDIO CLIPS	TARGET AUDIO FORMAT	VOICE PERSONA / TONE
23 Surgical Step Voiceover Tracks	MP3 (44.1 kHz / 48 kHz, Stereo or Mono, 192–320 kbps)	Preceptor Surgeon (Calm, authoritative, clear, 130–145 wpm)

 **Medical Pronunciation & TTS Direction Guide:**

- Phacoemulsification**
: *FAY-co-ee-mul-sih-fih-KAY-shun* (shortened to *FAY-co*)
- Paracentesis**
: *pair-uh-sen-TEE-sis* (small side-port corneal entry)
- Capsulorhexis (CCC)**
: *cap-syoo-lo-REK-sis* (continuous circular tear of lens capsule)
- Viscoelastic (OVD)**
: *vis-co-ee-LAS-tik* (clear protective jelly: *VIS-coat* / *PRO-visk*)
- Nd:YAG Laser**
: *EN-dee YAG* (Neodymium-doped Yttrium Aluminum Garnet)
- He-Ne Beams**
: *HEE-nee* (Helium-Neon red aiming laser)
- MIGS**
: *MIGZ* (Minimally Invasive Glaucoma Surgery)
- Trabecular Meshwork**
: *truh-BEK-yoo-ler MESH-werk*
- Schlemm's Canal**
: *SHLEMZ kuh-NAL*
- Schwalbe's Line**
: *SHVAL-bays line*

Step #	Target MP3 File	Surgical Step Title	Words	Est. Duration
PHACO 1	cataract_01_paracentesis.mp3	Paracentesis Incision (~1.0mm MVR)	51	~23 sec
PHACO 2	cataract_02_clear_corneal_incision.mp3	Tri-Planar Clear Corneal Incision (2.4mm)	56	~25 sec
PHACO 3	cataract_03_ovd_injection.mp3	OVD Injection (Dispersive & Cohesive)	41	~18 sec
PHACO 4	cataract_04_capsulorhexis.mp3	Continuous Curvilinear Capsulorhexis (CCC)	55	~24 sec
PHACO 5	cataract_05_hydrodissection.mp3	Hydrodissection & Free Rotation Test	55	~24 sec
PHACO 6	cataract_06_phaco_chop.mp3	Phaco-Chop Nucleofractis & Aspiration	63	~28 sec
PHACO 7	cataract_07_cortex_removal.mp3	Cortical Remnant Clearance (I/A)	46	~20 sec
IOL 8	cataract_08_ovd_bag_refill.mp3	Capsular Bag Refill with Cohesive OVD	43	~19 sec
IOL 9	cataract_09_cartridge_insertion.mp3	Cartridge Delivery into Bag	37	~16 sec
IOL 10	cataract_10_haptic_unfolding.mp3	Leading Haptic Placement	38	~17 sec
IOL 11	cataract_11_sinskey_dialing.mp3	Sinskey Hook 360° Rotational Centering	48	~21 sec
IOL 12	cataract_12_viscoelastic_washout.mp3	Retro-Lens & AC Viscoelastic Washout	50	~22 sec
YAG 1	yag_01_contact_lens_placement.mp3	Abraham Capsulotomy Lens Placement	49	~22 sec
YAG 2	yag_02_aiming_focus.mp3	Dual HeNe Aiming Beam Convergence	47	~21 sec
YAG 3	yag_03_offset_adjustment.mp3	Posterior Focal Offset (+150µm)	46	~20 sec
YAG 4	yag_04_cruciate_capsulotomy.mp3	Cruciate Pattern Laser Breakdown	54	~24 sec
YAG 5	yag_05_post_yag_assessment.mp3	Visual Axis Clearance & IOP Check	45	~20 sec
MIGS 1	migs_01_microscope_and_head_tilt.mp3	Operating Microscope & Patient Head Tilt	66	~29 sec
MIGS 2	migs_02_gonioprism_placement.mp3	Surgical Gonioprism Lens Placement	48	~21 sec
MIGS 3	migs_03_viscoelastic_angle_deepening.mp3	Viscoelastic Angle Deepening	41	~18 sec

Step #	Target MP3 File	Surgical Step Title	Words	Est. Duration
MIGS 4	migs_04_stent_1_deployment.mp3	Micro-Stent 1 Insertion into Schlemm Canal	57	~25 sec
MIGS 5	migs_05_stent_2_deployment.mp3	Micro-Stent 2 Insertion (Bilateral Bypass)	50	~22 sec
MIGS 6	migs_06_blood_reflux_and_washout.mp3	Episcleral Blood Reflux & Viscoelastic Washout	68	~30 sec

Procedure 1: Cataract Phacoemulsification & Foldable IOL (12 Steps)

STEP 1 Paracentesis Incision (~1.0mm MVR)

[cataract_01_paracentesis.mp3](#)

 EXACT SPOKEN SCRIPT (51 WORDS • ~23S):

"Step 1: Make a small side door. Take the 1.0 millimeter blade from the left tray. Look for the flashing orange target at the top-left edge of the eye, at the 10 o'clock position. Click there to make a tiny slit. This gives your assistant tool a way into the eye."

Recommended Instrument: MVR BLADE

Target Location: Upper-left edge of the cornea at 10:00 o'clock (limbus border between clear cornea and white sclera).

Action Guide:  Select the 1.0mm MVR Blade on the left tray, then CLICK the pulsing orange target at the 10 o'clock corneal edge.

Why Necessary: Eye surgery requires two hands. Your right hand holds the main ultrasound tool through the big incision, but you need a second tool (like a spatula or chopper) in your left hand to hold the cataract still, push pieces, and protect the back of the eye. Without this side door, you cannot control the cataract pieces and risk tearing the eye.

STEP 2 Tri-Planar Clear Corneal Incision (2.4mm)

cataract_02_clear_corneal_incision.mp3

EXACT SPOKEN SCRIPT (56 WORDS • ~25S):

"Step 2: Create your main doorway. Switch to the 2.4 millimeter keratome blade. Look for the flashing blue target at the upper-right, at the 1:30 o'clock position. Click 3 times to step through the three planes: groove the surface, tunnel through the wall, and enter the eye chamber. This creates a self-sealing tunnel that won't leak."

Recommended Instrument: KERATOME 2_4**Target Location:** Upper-right corneal edge at 1:30 o'clock, where clear cornea transitions into the white of the eye.**Action Guide:** 🖱️ Pick the 2.4mm Keratome blade. CLICK the pulsing blue target at the 1:30 o'clock position 3 times to complete Plane 1 (Groove), Plane 2 (Tunnel), and Plane 3 (AC Entry).**Why Necessary:** Your primary ultrasound pen (phaco handpiece) is 2.4mm thick. You must create an entry tunnel wide enough for the pen to slide through easily, yet tight enough that fluid doesn't rush out and collapse the eye. Cutting it in 3 staggered planes creates an automatic valve: inside eye pressure presses the inner flap shut, keeping the eye pressurized and preventing bacteria from entering post-surgery.**STEP 3 OVD Injection (Dispersive & Cohesive)**

cataract_03_ovd_injection.mp3

EXACT SPOKEN SCRIPT (41 WORDS • ~18S):

"Step 3: Protect the eye with jelly. Choose the Viscoat syringe on the left. Click inside the pupil to inject a clear protective gel. This coats the fragile inner lining of the cornea so ultrasonic soundwaves and turbulence won't damage it."

Recommended Instrument: OVD VISCOAT**Target Location:** Center of the pupil and the dome directly behind the cornea (the anterior chamber).**Action Guide:** 🖱️ Select Viscoat (dispersive jelly) or Provisc on the left tray, then CLICK inside the front chamber of the eye to inject the protective cushion.**Why Necessary:** The back surface of your cornea is lined with fragile "endothelial cells" that act like microscopic bilge pumps keeping the cornea clear. Unlike skin or hair, these cells NEVER grow back once lost! If ultrasound vibrates near them or turbulent water washes over them, they die, turning the cornea permanently milky white. The viscous jelly acts like a protective suit of armor and keeps the eyeball round and firm.

STEP 4 Continuous Curvilinear Capsulorhexis (CCC)

cataract_04_capsulorhexis.mp3

EXACT SPOKEN SCRIPT (55 WORDS • ~24S):

"Step 4: Cut a circular window in the lens skin. Take the needle cystotome to poke a small tear in the center, then use the Utrata micro-forceps to steer the flap around the dashed blue circle. Make a smooth 5.2 millimeter circle. If it starts running wild toward the edge, pull back toward the center."

Recommended Instrument: UTRATA FORCEPS**Target Location:** Center of the dark pupil, following the 5.2mm glowing blue circle guide.**Action Guide:** 🖱️ 1. Pick Cystotome and CLICK the center crosshair to puncture the skin. 2. Grab Utrata Forceps and DRAG in a smooth circle along the dashed blue guide ring.**Why Necessary:** If you just randomly poke or shred the bag, sharp tears will zip down the sides to the back of the eye like a run in a nylon stocking! If that happens, the entire cataract falls into the back of the eyeball (vitreous) and you cannot put a replacement lens in place. A smooth, continuous circle distributes tension evenly and has immense structural strength.**STEP 5 Hydrodissection & Free Rotation Test**

cataract_05_hydrodissection.mp3

EXACT SPOKEN SCRIPT (55 WORDS • ~24S):

"Step 5: Loosen the cataract with a fluid wave. Select the hydrodissection cannula. Slide the flat tip gently under the edge of your circular opening and click to spray a gentle pulse of balanced salt water. Look for the golden wave rolling across the back. Then verify the cataract spins freely like a dinner plate."

Recommended Instrument: HYDRO CANNULA**Target Location:** Underneath the anterior capsular rim at the top or side of your circular window.**Action Guide:** 🖱️ Pick the Hydro Cannula on the left tray, then CLICK just under the edge of the circular opening to inject the fluid wave.**Why Necessary:** The cataract is naturally glued to its protective bag by thousands of sticky cellular fibers. If you try to chop or turn the cataract while it is glued down, you will rip the fragile micro-threads (zonules) holding the bag to the eye wall, dropping the whole lens into the back of the eye! Water cleaves these bonds safely with zero mechanical stress.

STEP 6 Phaco-Chop Nucleofractis & Aspiration

cataract_06_phaco_chop.mp3

EXACT SPOKEN SCRIPT (63 WORDS • ~28S):

"Step 6: Pulverize and vacuum the hard cataract. Select the phaco tip. Step on your foot pedal to Position 3 to engage ultrasonic vibration. Touch the tip to the center of the brown lens core to impale it, chop it into smaller bite-sized quarters, and vacuum them up. Keep your tip in the middle and stay far away from the thin back capsule."

Recommended Instrument: PHACO TIP**Target Location:** Center of the pupil in the safe "phaco zone" (iris plane, pupil center, > 1.5mm away from the delicate back capsule).**Action Guide:** 🖐️ Pick the Phaco Tip. Click Foot Pedal to Position 3 (Ultrasound). CLICK and HOLD the central cataract core to chop and suck up all 4 quadrants.**Why Necessary:** A cataract is a rock-hard, cloudy natural lens up to 10 millimeters wide. You cannot pull a 10mm hard stone through a tiny 2.4mm keyhole incision without tearing the eye open! The phaco ultrasound needle turns the rock into soup (emulsification) so it fits right through the microscopic straw.**STEP 7 Cortical Remnant Clearance (I/A)**

cataract_07_cortex_removal.mp3

EXACT SPOKEN SCRIPT (46 WORDS • ~20S):

"Step 7: Vacuum the soft sticky leftovers. Switch to the Irrigation and Aspiration handpiece. Step on pedal Position 2 to turn on suction. Click around the outer edges to vacuum away the fluffy cortical fibers. Polish the back surface until it looks like a clean window."

Recommended Instrument: IA HANDPIECE**Target Location:** Equatorial periphery of the capsular bag (all 360 degrees around the outer rim).**Action Guide:** 🖐️ Pick the I/A Handpiece on the tray. Put Foot Pedal in Position 2 (Aspiration). CLICK around the periphery to vacuum all remaining cortical fluff.**Why Necessary:** Even after the hard core is gone, sticky cotton-candy-like material (cortex) clings to the bag walls. If you leave these cellular remnants inside, they cause severe eye inflammation, sky-high eye pressure (glaucoma), and trigger cloudy scar tissue that blinds the new implant.

STEP 8 Capsular Bag Refill with Cohesive OVD

cataract_08_ovd_bag_refill.mp3

 EXACT SPOKEN SCRIPT (43 WORDS • ~19S):

"Step 8: Re-inflate the empty bag. Select the Provisc cohesive jelly. Click inside the capsular bag to pump it full of thick jelly. This opens the bag wide so the new lens can slide in safely without poking a hole in the back."

Recommended Instrument: OVD PROVISC**Target Location:** Inside the circular capsulorhexis opening, aiming toward the back and corners of the bag.**Action Guide:**  Select Provisc (cohesive OVD) on the left tray, then CLICK inside the pupil to inflate the capsular bag.**Why Necessary:** Now that the cataract is gone, the clear bag collapses flat like an empty balloon. If you shove a hard plastic injector nozzle into a collapsed bag, it will spear right through the paper-thin back membrane! Injecting thick jelly acts like a balloon inflator, opening up a safe 3D pocket for the lens to unpack.**STEP 9 Cartridge Delivery into Bag**

cataract_09_cartridge_insertion.mp3

 EXACT SPOKEN SCRIPT (37 WORDS • ~16S):

"Step 9: Deliver the folded lens. Select the IOL injector. Slide the tapered nozzle through your main incision with the bevel pointing down. Click to smoothly advance the screw plunger and push the folded artificial lens inside."

Recommended Instrument: IOL INJECTOR**Target Location:** The 2.4mm main corneal incision at 1:30 o'clock, aiming towards the center of the pupil.**Action Guide:**  Select the IOL Injector on the left tray. CLICK on the main incision target to insert the nozzle and begin advancing the lens.**Why Necessary:** An artificial lens is 6 millimeters wide, but our incision is only 2.4 millimeters! The lens is folded up tightly like a microscopic burrito inside a sterile lubricated cartridge. The injector squeezes it through the tiny incision without stretching or tearing the corneal wound.

STEP 10 Leading Haptic Placement

cataract_10_haptic_unfolding.mp3

EXACT SPOKEN SCRIPT (38 WORDS • ~17S):

"Step 10: Place the front leg. Watch the leading springy arm unfold out of the nozzle directly into the far corner of the bag. Keep the nozzle steady in the center so the lens optic unrolls smoothly flat."

Recommended Instrument: IOL INJECTOR**Target Location:** Far (distal) corner equator of the capsular bag, opposite your incision.**Action Guide:** 🖱️ CLICK again with the IOL Injector to push the front arm and central optic fully out into the pupil plane.**Why Necessary:** If the spring arms unfold on top of the iris (the colored part of the eye) instead of inside the bag, they will rub against blood vessels, causing bleeding (hyphema), intense inflammation, and vision distortion. Both arms must seat neatly inside the bag pockets.**STEP 11** Sinskey Hook 360° Rotational Centering

cataract_11_sinskey_dialing.mp3

EXACT SPOKEN SCRIPT (48 WORDS • ~21S):

"Step 11: Tuck the back leg in. Select the tiny Sinskey hook tool. Hook into the corner notch of the lens and rotate it clockwise. Tuck the trailing arm under the edge of the circular opening. Check that the round opening overlaps the lens edge all 360 degrees."

Recommended Instrument: SINSKEY HOOK**Target Location:** Near the junction of the central optic and the trailing spring arm (proximal haptic).**Action Guide:** 🖱️ Select the Sinskey Hook on the tray. CLICK the edge of the lens to rotate it clockwise and tuck the trailing haptic inside the bag.**Why Necessary:** If one arm is inside the bag and the other is outside in the ciliary sulcus (asymmetric placement), the lens will tilt, causing severe blurred vision, double vision, and chronic eye pressure spikes. Furthermore, having the capsulorhexis overlap the lens 360 degrees locks it in place forever.

STEP 12 Retro-Lens & AC Viscoelastic Washout

cataract_12_viscoelastic_washout.mp3

 EXACT SPOKEN SCRIPT (50 WORDS • ~22S):

"Step 12: Vacuum out all the remaining jelly. Take the I/A suction handpiece. Gently tilt the lens and reach behind it to suck out all the thick jelly trapped in the back. If you leave even a little jelly behind, the patient will wake up with dangerously high eye pressure."

Recommended Instrument: IA HANDPIECE**Target Location:** Behind the new acrylic lens optic (the retro-lens space) and inside the front chamber angles.**Action Guide:** 🖱️ Select the I/A Handpiece. Put Foot Pedal in Position 2. CLICK behind the lens optic to vacuum out all residual viscoelastic jelly.**Why Necessary:** The surgical jelly that saved the cornea earlier will now destroy the eye if left behind! The microscopic drains of the eye (trabecular meshwork) get completely clogged by thick jelly. Fluid cannot drain out, causing intraocular pressure to skyrocket past 40 or 50 mmHg within hours of surgery, leading to agonizing pain, nausea, and irreversible optic nerve damage (glaucoma).

Procedure 2: Nd:YAG Laser Photodisruption Capsulotomy (5 Steps)

STEP 1 Abraham Capsulotomy Lens Placement

📁 yag_01_contact_lens_placement.mp3

🔊 EXACT SPOKEN SCRIPT (49 WORDS • ~22S):

"Step 1 of YAG Laser: Place the special contact lens on the eye. Put a drop of clear gel on the Abraham contact lens and place it onto the patient's cornea. The magnifying button sharpens the laser beam into a tight focus point and keeps the patient from blinking."

Recommended Instrument: YAG LASER

Target Location: Cornea surface, centered over the pupil.

Action Guide: 🖱️ With the YAG Laser module active, CLICK on the eye to apply the Abraham contact lens.

Why Necessary: Months or years after cataract surgery, microscopic cells grow across the back bag like frosting on glass (a secondary cataract). We use an invisible infrared laser to blast a clear window through the haze. The Abraham contact lens widens the laser cone angle from 16° to 24°, concentrating energy strictly at the cloudy membrane while making it harmless to the cornea in front and the retina in the back.

STEP 2 Dual HeNe Aiming Beam Convergence

📁 yag_02_aiming_focus.mp3

🔊 EXACT SPOKEN SCRIPT (47 WORDS • ~21S):

"Step 2: Align the twin red aiming lasers. Move your slit-lamp joystick forward or backward until the two separate red dots merge into a single crisp red point right on the cloudy membrane. When the two dots become one, you know the laser is perfectly in focus."

Recommended Instrument: YAG LASER

Target Location: On the opacified posterior capsule behind the artificial lens.

Action Guide: 🖱️ Move your cursor over the hazy membrane until the twin red aiming dots converge into a single tight red reticle.

Why Necessary: The therapeutic laser beam is 1064nm infrared—completely invisible to the human eye! To show you where it will strike, the machine projects two red aiming beams from different angles. When the two red spots merge into one, the invisible laser is focused with pinpoint accuracy. If they are separated, the laser will fire in the wrong depth plane!

STEP 3 Posterior Focal Offset (+150µm)

yag_03_offset_adjustment.mp3

EXACT SPOKEN SCRIPT (46 WORDS • ~20S):

"Step 3: Crucial safety setting! Set the posterior offset to plus 150 micrometers. Never fire at zero offset! When the laser sparks, the mini shockwave expands forward. If you don't set a safety gap, the shockwave will pit, scratch, and crack the patient's expensive artificial lens!"

Recommended Instrument: YAG LASER**Target Location:** The right machine console: "Defocus Offset" dial set to +150 µm.**Action Guide:** 🖱️ Verify the Laser Defocus Offset is set between +150µm and +200µm in the right console panel.

Why Necessary: This is the number one rookie mistake in ophthalmology! When a laser spark creates plasma, the plasma sparks and expands FORWARD back toward the incoming laser beam. If your focus is set directly on the capsule (zero offset), the shockwave strikes the posterior surface of the plastic lens, blasting permanent micro-craters (pits) into the optic, creating glare, halos, and permanent vision distortion for the patient.

STEP 4 Cruciate Pattern Laser Breakdown

yag_04_cruciate_capsulotomy.mp3

EXACT SPOKEN SCRIPT (54 WORDS • ~24S):

"Step 4: Zap the membrane in a cross pattern. Fire your laser pulses starting at the outer edges: 12 o'clock at the top, 6 o'clock at the bottom, then 9 and 3 o'clock. Cutting in a cross releases the tension, making the cloudy flaps curl up and roll out of the visual axis naturally."

Recommended Instrument: YAG LASER**Target Location:** The 4 arms of the cross (+) in the hazy pupil area outside the central line of sight.**Action Guide:** 🖱️ CLICK on the numbered cross targets on the cloudy membrane: 1 (Top 12h) -> 2 (Bottom 6h) -> 3 (Left 9h) -> 4 (Right 3h).

Why Necessary: If you blast directly in the dead center first, any accidental lens pit will land right in the patient's central vision line! Furthermore, the cloudy capsule is under tight drum-skin tension. By cutting the edges first like cutting the ropes of a tent, the tension pulls the central flaps wide open automatically, clearing vision using minimum laser energy.

STEP 5 Visual Axis Clearance & IOP Check

yag_05_post_yag_assessment.mp3

 EXACT SPOKEN SCRIPT (45 WORDS • ~20S):

"Step 5: Inspect your work and check pressure. Verify you have created a clean 4 millimeter circular window with no loose tags hanging in the center. Check that the gel bag behind the eye is intact, and apply eye drops to prevent post-laser pressure spikes."

Recommended Instrument: YAG LASER**Target Location:** Central 4mm optical zone of the pupil and the anterior vitreous face.**Action Guide:** 👉 Verify the visual axis is clear, confirm 0 IOL pits, and finalize the procedure.**Why Necessary:** Loose tags hanging in the pupil will swing back and forth like a pendulum, creating ghostly double vision and annoying glare for the patient. Also, laser shockwaves release microscopic debris that can clog eye drains for 1 to 4 hours post-op, requiring prophylactic pressure-lowering drops (apraclonidine or brimonidine).

Procedure 3: MIGS Glaucoma Trabecular Micro-Bypass Stents (6 Steps)

STEP 1 Operating Microscope & Patient Head Tilt

migs_01_microscope_and_head_tilt.mp3

EXACT SPOKEN SCRIPT (66 WORDS • ~29S):

"Step 1 of MIGS Glaucoma Stent Surgery: Angle the microscope and patient head. Because the drainage angle of the eye is hidden around the side curve behind the cornea, light from straight above cannot see it. Tilt the microscope 35 to 40 degrees towards yourself, and tilt the patient's head 30 to 35 degrees away. Click the alignment beacon to set optimal direct gonioscopic optical trajectory."

Recommended Instrument: NONE

Target Location: Microscope ocular axis and patient cervical rotation angle.

Action Guide: 🖱️ CLICK the Tilt Alignment control on the angle HUD to align the microscope and patient head to 38°/35°.

Why Necessary: Normal light from a surgical microscope shines straight down into the pupil. But the eye's microscopic drainage canal (Schlemm's canal) is tucked sideways around the 360-degree perimeter of the corneal rim! If you look straight down, total internal reflection inside the curved cornea completely hides the drain! Tilting the microscope and head lets you peek sideways into the drainage angle.

STEP 2 Surgical Gonioprism Lens Placement

migs_02_gonioprism_placement.mp3

EXACT SPOKEN SCRIPT (48 WORDS • ~21S):

"Step 2: Apply the Swan-Jacob surgical gonioprism. Select the gonio lens from your tray. Place a drop of cohesive viscoelastic on the front surface of the cornea, then gently couple the flat lens onto the eye. Look for the golden-brown pigmented band: that is the diseased trabecular meshwork."

Recommended Instrument: GONIO LENS

Target Location: Anterior corneal surface and nasal iridocorneal angle.

Action Guide: 🖱️ Select the Swan-Jacob Gonioprism on the tray, then CLICK the center of the cornea to lock the high-definition angle view.

Why Necessary: Even with tilt, the curved cornea bends light like a dome mirror. A surgical gonioprism is an optical glass wedge that cancels the optical curvature of the cornea. It turns the curved surface into an optically flat window, revealing the four key landmarks: Schwalbe's line, the pigmented trabecular meshwork, the scleral spur, and the ciliary body band.

STEP 3 Viscoelastic Angle Deepening

migs_03_viscoelastic_angle_deepening.mp3

EXACT SPOKEN SCRIPT (41 WORDS • ~18S):

"Step 3: Deepen the drainage corner with thick cohesive jelly. Select Provisc on the tray. Inject a gentle bolus into the nasal angle. Watch the iris push backward away from the cornea, opening up a spacious cavern for the stent injector."

Recommended Instrument: OVD PROVISC**Target Location:** Nasal anterior chamber angle recess between the peripheral cornea and iris root.**Action Guide:** 🖱️ Select Provisc on the tray, then CLICK the nasal angle target to expand the iridocorneal recess.

Why Necessary: In many glaucoma patients, the iris sits very close to the drainage meshwork (a crowded or narrow angle). If you attempt to slide an injector into a crowded angle, the sharp metal tip will gouge into the iris, causing bleeding (hyphema) and severe inflammation. Injecting cohesive jelly acts like an architectural pillar, pushing the iris back and giving you a wide, safe runway.

STEP 4 Micro-Stent 1 Insertion into Schlemm Canal

migs_04_stent_1_deployment.mp3

EXACT SPOKEN SCRIPT (57 WORDS • ~25S):

"Step 4: Deploy Micro-Stent number 1. Select the iStent inject pen. Guide the microscopic tip across the eye to the nasal golden-brown meshwork band at 2:30 o'clock. Approach at a 15 to 20 degree angle. Pierce through the meshwork right into Schlemm's canal and click the deployment button! The stent now channels fluid straight into the bloodstream."

Recommended Instrument: MIGS INJECTOR**Target Location:** Pigmented trabecular meshwork at 2:30 o'clock in the nasal quadrant, directly over a primary collector channel ostium.**Action Guide:** 🖱️ Select the iStent Injector Pen. CLICK the flashing green beacon at 2:30 o'clock on the pigmented meshwork to seat Micro-Stent 1.

Why Necessary: Here is the core secret: In glaucoma, 70% to 90% of the clog is in the trabecular meshwork filter itself! But the veins right behind it (Schlemm's canal and episcleral veins) are wide open and healthy! By punching this microscopic titanium snorkel directly through the clog, eye fluid bypasses the blockage completely and dumps directly into the bloodstream! And because the venous bloodstream has an automatic backpressure of 8 to 10 mmHg, the eye can never over-drain or collapse!

STEP 5 Micro-Stent 2 Insertion (Bilateral Bypass)

migs_05_stent_2_deployment.mp3

EXACT SPOKEN SCRIPT (50 WORDS • ~22S):

"Step 5: Deploy the second micro-stent. Retract the injector tip slightly, slide 2 clock hours down to 4:00 o'clock, and target the adjacent collector channel. Press gently into the pigmented meshwork and click to deploy Stent 2. Having two stents doubles your outflow capacity and guarantees dramatic eye pressure reduction."

Recommended Instrument: MIGS INJECTOR**Target Location:** Pigmented trabecular meshwork at 4:00 o'clock in the nasal-inferior quadrant (~2.0mm distance from Stent 1).**Action Guide:** 🖱️ With the iStent Injector Pen still selected, CLICK the flashing cyan beacon at 4:00 o'clock to deploy Micro-Stent 2.**Why Necessary:** Schlemm's canal has discrete outflow collector channels spaced around the eye like highway exit ramps. Placing two micro-stents spaced 2 clock hours (~2 mm) apart taps into multiple collector channel networks across the nasal quadrant. This drops eye pressure twice as effectively and provides backup insurance in case one channel ever scars down.**STEP 6 Episcleral Blood Reflux & Viscoelastic Washout**

migs_06_blood_reflux_and_washout.mp3

EXACT SPOKEN SCRIPT (68 WORDS • ~30S):

"Step 6: Verify blood reflux and clean the eye. Take the I/A suction handpiece. Gently tap the corneal wound to let a tiny drop of fluid out. Watch the miraculous blood reflux wave! Bright red blood seeps backwards through both stents from the bloodstream into the eye. This proves your stents are connected directly to the bloodstream! Wash out all remaining jelly, and your glaucoma surgery is complete."

Recommended Instrument: IA HANDPIECE**Target Location:** Nasal trabecular meshwork stent lumens and anterior chamber angle.**Action Guide:** 🖱️ Select the I/A Handpiece, put Foot Pedal in Position 2. CLICK to confirm the blood reflux wave and vacuum out all viscoelastic jelly.**Why Necessary:** The Blood Reflux Wave is the definitive holy grail test in glaucoma surgery! When you temporarily lower the eye pressure slightly below 8 to 10 mmHg, venous blood from the body's bloodstream flows backward out of Schlemm's canal into the stent inlet! Seeing that bright red blood plume proves 100% that your stents are in the bloodstream! And because the blood pressure will always push back at 8 to 10 mmHg, the patient's eye pressure can never drop too low (protecting against hypotony)!